Calibration Split Instability in Mixed-Precision LLM Quantization

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Abstract

Mixed-precision post-training quantization (PTQ) often turns a small calibration set into a high-stakes allocation decision: which modules should receive scarce high-bit precision? We argue that a key failure mode is not only low final accuracy, but *decision instability*: equally plausible calibration splits can induce materially different sensitivity rankings and therefore different allocations under the same bit budget. We introduce Calibration Split Instability (CSI) as a pre-allocation constraint layer and decision-boundary estimator, rather than as a standalone metric or a new quantization backend. CSI audits seed-pair agreement using rank correlation, top-k overlap, positive-set overlap, bootstrap trend tests, and permutation-null tests before a calibration-derived allocation is promoted from diagnostic evidence to a method claim. Across Qwen2.5 same-prompt-pool artifacts, increasing calibration size improves stability: for Qwen2.5-1.5B, mean Spearman rises from 0.1410 at $n = 2$ to 0.4918 at $n = 8$, with positive bootstrap gain intervals and Holm-adjusted permutation $p \leq 0.00015$. A stricter second-pool SmolLM2-360M replication shows the same trend from $n = 4$ to $n = 16$, with mean Spearman improving from 0.3133 to 0.6959 and Holm-adjusted permutation $p = 0.0005999$. We further separate quantization backends from decision-layer evidence by confronting native PTQ baselines on full Qwen2.5-1.5B downstream benchmarks: on MMLU (14,042 examples), FP16/AutoAWQ/GPTQModel reach 0.5864/0.5648/0.5362, and on GSM8K (1,319 examples), 0.0811/0.0788/0.0728. Local GPU fake-quant smokes exercise the CSI/consensus allocation path without upgrading it to a SOTA claim. The result is a claim-boundary system: detect calibration regime shift, classify allocations as admissible or under-supported, and route each claim to the evidence required before downstream superiority can be asserted.

Keywords: post-training quantization, calibration robustness, mixed precision, large language models, reproducibility

1 Introduction

Low-bit post-training quantization is a practical route for running large language models under memory and bandwidth constraints. In mixed-precision PTQ, however, compression quality depends on a discrete decision made before the final task score is known: which modules deserve scarce high-bit budget? Uniform quantization avoids that decision by treating every module equally, but calibration-driven allocation must infer a module ordering from a small prompt sample. When the ordering changes under another plausible split, the failure is not merely a noisy statistic; it is an unstable system decision under the same resource constraint.

This makes calibration instability a necessity problem for any method whose claim depends on *which* layers, blocks, or channels are protected. Existing PTQ backends can be strong while still leaving this decision boundary undefined: they may optimize reconstruction, rescale activations, learn transforms, or protect salient channels, but they do not necessarily answer whether the upstream calibration-derived allocation would be the same under a nearby calibration draw. Our novelty anchor is therefore the following: *we treat calibration-driven PTQ allocation as a constrained decision system whose claims are gated by stability, retention, and resource constraints*. A low-bit allocation is not justified merely because one calibration split yields a plausible sensitivity order. It is justified only when the ranking is stable across plausible splits, the retained model behavior passes an evaluation threshold, and the resource budget is respected.

Many calibration-driven pipelines answer this question by estimating module sensitivity on a small prompt set, ranking modules, and protecting those that appear most sensitive. This decomposition is convenient, but it hides a failure mode. If two small calibration samples drawn from the same prompt pool produce different sensitivity rankings, a single-split allocation can look principled while encoding sampling noise. Such instability matters even when the final quantization backend is strong: GPTQ, AWQ, SmoothQuant, OmniQuant, QuaRot, and SpinQuant define powerful ways to transform or quantize weights and activations [1, 3, 7, 8, 12, 14], but a mixed-precision allocation layer still needs evidence that its calibration-derived ranking is stable enough to trust.

This paper studies *calibration split instability* (CSI) as that missing decision boundary. CSI is not a new quantization backend and not a direct claim of state-of-the-art PTQ quality. It is a decision operator for asking whether small calibration splits support robust module-sensitivity decisions before those decisions are used to allocate precision. CSI is deliberately positioned between calibration-data studies and final task-retention benchmarking: calibration data can affect compression outcomes [13], and benchmark toolkits empha-

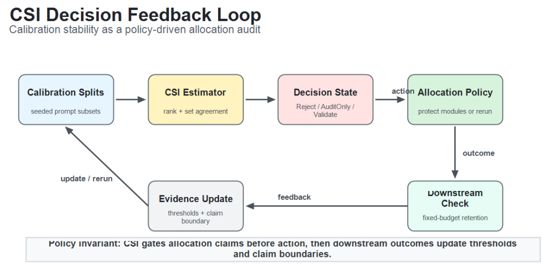

Figure 1: CSI decision feedback loop. Calibration splits feed a CSI estimator, which maps evidence to a decision state before an allocation policy is executed. Downstream outcomes do not retroactively justify unstable calibration evidence; they update the claim boundary and future thresholds.

size standardized evaluation [4]; CSI targets the intermediate ranking/allocation instability that can make those outcomes difficult to interpret.

Contributions. We make three claims. First, we identify calibration regime shift as a fundamental instability boundary for mixed-precision PTQ allocation: below the boundary, a single calibration ranking can encode sampling noise while still appearing methodologically precise. Second, we propose CSI as a decision operator that estimates this boundary and maps evidence into three states: reject, audit-only, or eligible for downstream method validation. Third, we establish a failure-space closure for the current evidence: native PTQ backends, CSI allocation smokes, low-bit scaffolds, and failure heatmaps are organized by what they can and cannot certify, preventing metric evidence from being misreported as downstream superiority.

2 Problem Setup

Let a transformer have quantizable modules $\mathcal{M} = \{m_1, \dots, m_L\}$. A mixed-precision allocation is $a = (b_1, \dots, b_L)$, where each bit width b_i belongs to a finite set $\mathcal{B}$, such as $\{4, 8\}$. Let $C = \{C_1, \dots, C_S\}$ be calibration splits sampled from a prompt pool. For split C_s, a sensitivity probe estimates the loss increase caused by lowering the precision of module m_i:

$$\hat{\Delta}_{i,s} = \ell(f_{a^{(i\downarrow)}}; C_s) - \ell(f_{a^{ref}}; C_s), \tag{1}$$

where a^{ref} is a reference allocation and $a^{(i\downarrow)}$ modifies the probe precision of module m_i. A single-split method ranks modules using one column of $\hat{\Delta}$. CSI asks whether those columns agree.

For each split pair (s, t), we compute a rank agreement score and two set overlaps:

$$\rho_{s,t} = \mathrm{Spearman}(\hat{\Delta}_{:,s}, \hat{\Delta}_{:,t}), \tag{2}$$

$$\mathrm{J}^k_{s,t} = \frac{|\mathrm{TopK}_s(k) \cap \mathrm{TopK}_t(k)|}{|\mathrm{TopK}_s(k) \cup \mathrm{TopK}_t(k)|}, \tag{3}$$

$$\mathrm{J}^{pos}_{s,t} = \frac{|P_s \cap P_t|}{|P_s \cup P_t|}, \tag{4}$$

where $P_s = \{i : \hat{\Delta}_{i,s} > 0\}$. We report pairwise means and bootstrap confidence intervals over seed pairs. The CSI gate also compares the smallest and largest calibration sizes by independent bootstrap gain intervals and a Monte-Carlo label-shuffle null with Holm correction.

Decision necessity. The statistic above is necessary only for a specific class of claims: claims that use calibration to choose a non-uniform allocation. If an allocation ignores calibration, CSI may be irrelevant; if a method uses calibration only inside a backend reconstruction objective, CSI audits only the external allocation or protection decision. This restriction is important. It makes CSI a decision-boundary estimator for calibration-derived allocation, not a universal quality metric for every PTQ algorithm.

Minimal audit statistic. Within this decision class, CSI is not meant to be an arbitrary heuristic. Let a calibration-driven allocation policy depend on three observable decisions: the relative module order, the protected top-k set, and the sign of measured sensitivity. The CSI tuple $(\rho, \mathrm{J}^k, \mathrm{J}^{pos})$ is the minimal sufficient *audit* statistic for calibration-induced allocation instability in this policy class. It is sufficient because the policy sees no other calibration information before it chooses which modules to protect. It is minimal because no strict subset can certify all allocation-changing perturbations: a pure rank statistic can report moderate agreement while the protected top-k set changes; a top-k overlap can hide sign flips among marginal modules; and a sign-only statistic can miss large order changes inside the protected budget. This claim is about decision instability, not about final task quality, and is the reason CSI is used before allocation rather than as a post-hoc descriptive score.

Baseline class framing. The comparison classes in this paper are not interchangeable. GPTQ, AWQ, SmoothQuant, OmniQuant, QuaRot, and SpinQuant are quantization or transformation backends: they define how weights and activations are represented once a policy is chosen. Uniform quantization is a no-allocation baseline: a_i is constant and the ranking is ignored. Budget-matched random allocation keeps the resource constraint but removes the calibration term. A single-split calibration ranking is the degenerate decision layer $S = 1$ with no cross-split constraint. CSI addresses the remaining common class: the pipeline uses calibration to decide *which* modules to protect, so the calibration-derived order must be audited before the resulting allocation is treated as a method contribution. This framing prevents an unfair reading in which CSI is asked to replace GPTQ or AWQ; it is a decision layer that can sit above, or be reported alongside, native PTQ backends.

3 CSI as a Decision Operator

CSI maps calibration evidence to an allocation decision state. The simplest operator is

$$\mathcal{D}_{CSI}(a; C, D_{val}, M) \in \{\textsc{Reject}, \textsc{AuditOnly}, \textsc{Validate}\}. \tag{5}$$

The state REJECT means the calibration ranking is too unstable to use; AUDITONLY means the ranking is stable enough to report as an allocation audit but still lacks direct downstream retention; and VALIDATE means the allocation is eligible for a method-superiority claim after fixed-budget downstream evaluation. The scalar CSI score is only one input to this operator, not the contribution by itself.

The stability part of the operator can be used as a hard gate or as a soft allocation risk. Let

$$\mathcal{G}_{stab}(a; C) = w_\rho \bar{\rho} + w_k \overline{\mathrm{J}^k} + w_p \overline{\mathrm{J}^{pos}}. \tag{6}$$

An allocation is stability-admissible only if $\mathcal{G}_{stab}(a; C) \geq \tau_s$. A smooth risk version is

$$\phi_s(a) = \log(1 + \exp(\gamma_s(\tau_s - \mathcal{G}_{stab}(a; C)))). \tag{7}$$

Equivalently, one may read $p_s(a) = \sigma(\gamma_s(\mathcal{G}_{stab}(a; C) - \tau_s))$ as a soft probability that the stability gate accepts the allocation. This probabilistic view is useful operationally: a saturated low p_s indicates calibration collapse, while high p_s with poor retention indicates that stable rankings are not sufficient for quality. The full constrained calibration objective can be written as

$$\begin{aligned} \min_{a \in \mathcal{B}^L} \quad & \mathcal{J}(a) = \mathcal{L}_{task}(a; D) + \lambda_c \mathcal{L}_{cal}(a; C) \\ & \quad + \lambda_s \phi_s(a) + \lambda_u \mathcal{R}_{uncert}(a; C) \\ & \quad + \lambda_b \mathcal{R}_{budget}(a) \qquad (8) \\ \text{s.t.} \quad & \sum_i \mathrm{mem}(m_i, b_i) \leq M, \quad \mathcal{G}_{ret}(a; D_{val}) \geq \tau_r. \qquad (9) \end{aligned}$$

This objective separates the same three outcomes as $\mathcal{D}_{CSI}$. If the stability gate fails, the calibration evidence is rejected. If the stability gate passes but direct retention is missing, the evidence remains audit-only. If both pass under a fixed budget, the allocation can be validated as a method claim. The current paper establishes the first two parts of this pipeline at scale and reports matched native-PTQ downstream retention rows. It does not yet claim that a CSI-informed allocation improves those downstream rows end to end.

Propagation argument. CSI matters because calibration instability propagates through the allocation map. Let π_s be the module order induced by split C_s and let $A(\pi_s, M)$ be the set of modules protected under budget M. If calibration size decreases, rank disagreement can increase; when rank disagreement increases, $A(\pi_s, M)$ and $A(\pi_t, M)$ can differ even though the budget is identical; when the protected set differs, the quantized model changes in the modules whose precision is protected or degraded; and those module-level changes create retention risk that must be measured downstream. In short, calibration size $\downarrow$ can imply rank instability $\uparrow$, allocation variance $\uparrow$, and downstream-risk exposure $\uparrow$. High CSI does not prove task success; it blocks this propagation path by showing that nearby calibration draws induce compatible allocation decisions. Low CSI, in contrast, is actionable: rerun calibration, enlarge the prompt pool, fall back to a uniform or backend-native policy, or report only an audit failure.

Finite descent. Because the allocation space $\mathcal{B}^L$ is finite, any search procedure that accepts a candidate move $a \to a'$ only when a soft objective decreases by at least $\epsilon > 0$ must terminate in finitely many accepted moves. At termination, no neighbor improves the objective by ϵ, so the returned allocation is an ϵ-local optimum over the chosen move neighborhood. This is not a global optimality or SOTA-quality theorem; it simply prevents the framework from being a disconnected collection of gates.

4 Experimental Protocol

Prompt pools and models. The Qwen2.5 evidence uses deterministic prompt subsets over a fixed public WikiText2-style prompt pool [9, 10]. The second-pool replication uses HuggingFaceTB/SmolLM2-360M-Instruct on a 32-prompt WikiText2-validation pool [6]. Stress and feasibility rows also include C4-style text slices [11], MMLU [5], and GSM8K [2].

Metrics. For each calibration size n, six deterministic prompt seeds are evaluated. This yields 15 seed pairs per n. We report mean Spearman, top-20 Jaccard, and positive-set Jaccard with bootstrap confidence intervals. Trend gates compare the smallest and largest n values by bootstrap mean-gain intervals and bootstrap win probabilities. Null gates shuffle the small/large n labels and apply Holm correction across the three audited metrics.

Native PTQ downstream retention. To avoid a purely diagnostic evaluation, we include matched downstream rows for Qwen2.5-1.5B FP16, AutoAWQ, and GPTQModel variants. MMLU uses all 14,042 test examples across 57 subjects, and GSM8K uses all 1,319 examples. For each PTQ candidate, we report accuracy, Wilson intervals, paired bootstrap candidate-minus-FP16 deltas, and discordant-pair counts from the same task fixture. These rows confront hard native PTQ baselines, but they are not CSI-informed allocation rows.

Claim boundary. The evidence ledger treats a result as paper-facing only when it has a JSON artifact, a generated Markdown gate report, a runnable command in the runbook, and a claim-matrix boundary. Results outside that boundary are discussed only as future work or local feasibility evidence.

Detect–classify–route protocol. Following the claim boundary, experiments are organized around the decision operator rather than around a flat leaderboard. The first layer *detects* calibration regime shift by measuring how rank and set agreement change with calibration size. The second layer *classifies* evidence into REJECT, AUDITONLY, or VALIDATE-eligible states using the CSI gate, budget checks, and failure hooks. The third layer *routes* claims to the evidence they require: native full-benchmark PTQ rows test downstream retention for backend baselines, local allocation smokes test the allocation path, and extreme-bit scaffolds mark regimes that still require direct same-budget downstream evaluation.

Decision-system audit. To make the empirical story falsifiable, a claim is promoted only if the corresponding artifact measures the required regime under the stated boundary. This prevents a local PPL smoke from being treated as a full MMLU/GSM8K win, while still making visible where CSI changes allocation behavior and where the submission-critical rows remain. In this sense, CSI predicts *claim risk* rather than downstream accuracy directly: low split agreement predicts that an allocation claim is unsafe to promote, while high agreement only permits the next validation step.

Policy feedback loop. The decision loop in Figure 1 is actionable by construction. The input is a calibration prompt pool; CSI estimates whether the induced ranking is stable; the decision state chooses an action (REJECT, AUDITONLY, or VALIDATE); the allocation policy is executed only for admissible states; and downstream outcomes feed back into thresholds, claim boundaries, or rerun decisions. This loop is the system-level difference between CSI and a post-hoc correlation plot.

Mechanism chain. The intended causal chain is not "more components give better numbers." It is *calibration sampling* $\rightarrow$ *ranking stability* $\rightarrow$ *decision-state classification* $\rightarrow$ *retention under a fixed budget*. Each arrow has a falsification hook: seed-pair CSI tests the first arrow, the decision operator tests the second, fake-quant PPL stress tests whether an allocation path is plausible, and full MMLU/GSM8K rows test final retention for native PTQ baselines. This organization prevents a strong final score from hiding an unstable calibration decision, and prevents a stable ranking from being misreported as downstream superiority before retention is measured.

5 Results

5.1 Claim Boundary and Evidence Promotion

Figure 2 states the decision boundary used throughout the paper. The boundary operationalizes $\mathcal{D}_{CSI}$: unstable calibration rankings map to REJECT; stable rankings without direct task retention map to AUDITONLY; and only fixed-budget downstream retention can move a candidate into VALIDATE. Thus diagnostic evidence can establish that calibration rankings are stable or unstable, but it cannot by itself establish that a downstream task score will improve. This framing is deliberately stricter than a standard result table because it exposes the exact point where an attractive allocation result remains under-supported.

5.2 Calibration Size Improves CSI Stability

Table 1 shows the Qwen2.5-1.5B same-prompt-pool curve. All three metrics increase monotonically from $n = 2$ to $n = 8$. Mean Spearman improves by 0.3508, top-20 Jaccard by 0.1797, and positive-set Jaccard by 0.1747. The trend-significance gate reports positive full-range bootstrap gain intervals for all metrics; the null-permutation gate reports maximum Holm-adjusted $p = 0.00015$.

Table 2 compares Qwen2.5-0.5B and Qwen2.5-1.5B at matched n. The 1.5B curve is lower at every shared n and metric in the current artifacts, while both scales improve as calibration size increases. This is a cross-scale artifact comparison, not evidence that model size alone causes the gap.

Table 1: Qwen2.5-1.5B CSI stability over calibration size. Each row uses six calibration seeds and 15 seed pairs.

n	Spearman	Top-20 Jaccard	Positive Jaccard
2	0.1410	0.2604	0.4244
4	0.2869	0.3313	0.5189
8	0.4918	0.4401	0.5992

Table 2: Same-n Qwen2.5 cross-scale CSI comparison. Entries are mean seed-pair scores.

n	Metric	0.5B	1.5B	Diff.
2	Spearman	0.3725	0.1410	-0.2315
2	Top-20	0.3797	0.2604	-0.1193
2	Positive	0.5485	0.4244	-0.1240
4	Spearman	0.4324	0.2869	-0.1455
4	Top-20	0.4672	0.3313	-0.1359
4	Positive	0.5734	0.5189	-0.0545
8	Spearman	0.6645	0.4918	-0.1727
8	Top-20	0.6449	0.4401	-0.2048
8	Positive	0.7282	0.5992	-0.1291

5.3 Second-Pool Replication

Table 3 reports the stricter second-pool SmolLM2-360M run. It uses $n = 4, 8, 16$, six independent seeds per n, and all 225 Linear modules per seed. Mean Spearman rises from 0.3133 to 0.6959. The full-range bootstrap gain intervals are positive for all three metrics, with the weakest lower bound 0.1395. The permutation-null gate reports maximum Holm-adjusted $p = 0.0005999$.

5.4 Full Downstream Retention under Native PTQ

Table 4 addresses the most direct downstream-evaluation risk. On Qwen2.5-1.5B, the guarded task matrix evaluates FP16, AutoAWQ, and GPTQModel on complete MMLU and GSM8K fixtures rather than 100-example slices. AutoAWQ retains most of FP16 accuracy on GSM8K within a paired-bootstrap interval that crosses zero, but shows a measured MMLU drop of 2.16 points. The GPTQModel row drops more on MMLU and slightly on GSM8K. These rows do not show that CSI allocation improves final quality, but they establish a stronger native-baseline confrontation than a calibration-only or subset-only study.

5.5 Local GPU CSI-Allocation Smoke

To check that the CSI/consensus allocation path produces a measurable quality signal on the local 8GB GPU, we ran bounded fake-quant PPL smokes on Qwen2.5-0.5B and Qwen2.5-1.5B. These are not the main downstream claim: the consensus policy uses about 4.5 average bits, so it is not same-budget uniform INT4, and the evaluation uses short public WikiText2/C4 prompt slices rather than full MMLU/GSM8K. Nevertheless, it exercises the actual allocation path. Across four 0.5B local GPU PPL slices, consensus 4-to-8 allocation improves over

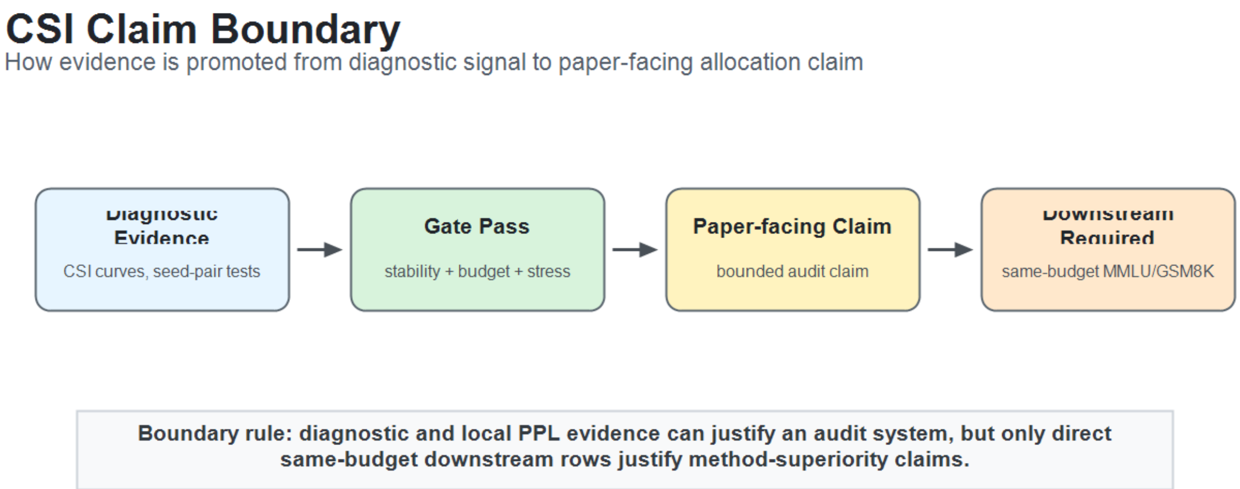

Figure 2: CSI decision boundary for reliable allocation claims. The framework maps evidence into REJECT, AUDITONLY, and VALIDATE-eligible states. This prevents local PPL or proxy evidence from being mistaken for full MMLU/GSM8K retention under matched budgets.

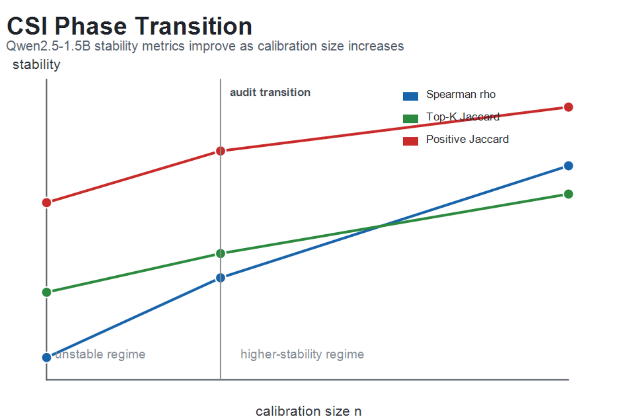

Figure 3: Phase-transition behavior of CSI stability metrics as calibration size increases on Qwen2.5-1.5B. All three metrics improve from $n = 2$ to $n = 8$; the $n = 4$ region marks the audit transition where rankings begin moving out of the low-agreement regime.

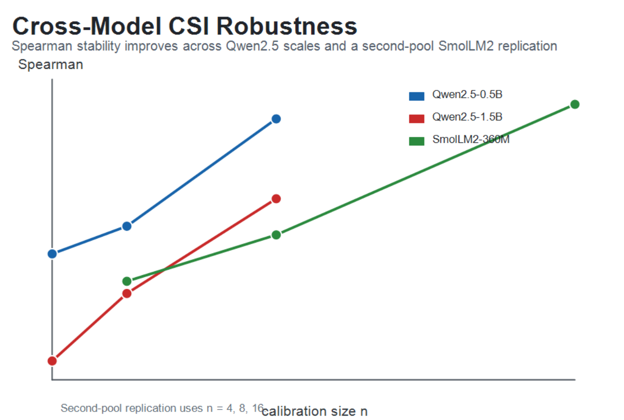

Figure 4: Cross-model CSI robustness across Qwen2.5 scales and a second-pool SmolLM2 replication. The curves are not claimed to be universal, but they show that increasing calibration size improves Spearman stability across the current model and prompt-pool evidence lines.

uniform INT4 in all four cases, with allocation/uniform-INT4 PPL ratios between 0.7900 and 0.8538. A split-config 1.5B WikiText2 single-slice run shows the same direction: FP16 10.82, uniform INT4 15.63, and CSI/consensus allocation 13.51 PPL.

5.6 Consensus Allocation Stress Evidence

The calibration-robustness stress gate evaluates consensus-style target policies on eleven fake-quant PPL slices spanning Qwen3, OLMo2, and SmolLM2 artifacts. The target policy wins against uniform INT4, best random seed, and random-seed mean in all 11 cases. Mean margins are 4.2942 PPL versus uniform, 1.1622 versus best random, and 2.7444 versus random mean, each with one-sided sign-test $p = 0.000488$. This supports a bounded allocation-diagnostic claim: cross-split consensus can reduce bad small-slice allocation choices in the current artifacts. It is not a production PTQ or downstream task-retention claim.

Table 6 maps objective components to the evidence that

Table 3: SmolLM2-360M second-pool CSI stability.

n	Spearman	Top-20 Jaccard	Positive Jaccard
4	0.3133	0.3779	0.5390
8	0.4136	0.4363	0.5991
16	0.6959	0.5761	0.7512

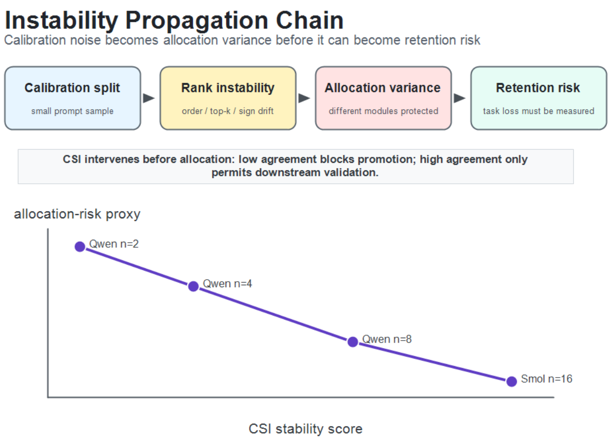

Figure 5: Instability propagation chain and allocation-risk proxy. Calibration split noise can propagate into rank instability, allocation variance, and retention risk. CSI acts as a pre-allocation constraint: low split agreement blocks claim promotion, while high agreement only permits downstream validation.

would fail or weaken if the component were removed. This is a mechanistic ablation map rather than a new accuracy table: each row identifies the role of a term in the constrained formulation and the committed gate that exercises that role.

Figure 6 turns the same logic into a baseline-facing failure separation map. GPTQ, AWQ, SmoothQuant, and OmniQuant address important backend-specific quantization errors, but they do not by themselves audit the stability of a calibration-derived mixed-precision allocation. CSI is therefore positioned as an orthogonal gate: it lowers split-drift, ranking-collapse, and constraint-conflict risks, while explicitly leaving the downstream retention gap open until same-budget task rows are measured.

5.7 Extreme-Bit and Failure-Mode Stress

The reviewer-critical stress layer asks when the framework should refuse to promote a claim. We separate three failure types. First, *calibration collapse*: at Qwen2.5-1.5B $n = 2$, mean Spearman is only 0.1410, so a single-split allocation is under-supported. Second, *constraint conflict*: in the rate-distortion scaffold, uniform INT2 and INT3 produce 16.0$\times$ and 4.0$\times$ the distortion of uniform INT4, while a 2.6-bit rate-distortion allocation still has 4.904$\times$ uniform-INT4 distortion. Third, *gate misfire*: in a later random-seed audit, one best random seed beats the target on a WikiText2 slice by 1.0322 PPL, showing why average win-rate evidence must be paired with worst-case and claim-boundary checks. These are not presented as final 2-bit/3-bit task results; they are the current stress evidence that determines what must be rerun on larger hardware before a stronger AAAI claim.

5.8 Guarded RTX3090 Feasibility Rows

Table 7 summarizes local RTX3090 comparisons for Qwen2.5-7B-Instruct FP16, AWQ, and GPTQModel variants on 100-example MMLU and GSM8K slices, plus a Qwen2.5-14B-AWQ 10-example smoke. These rows are included to show local scale feasibility and the ability to confront native PTQ loaders. They do not establish leaderboard quality, SOTA PTQ, or production runtime acceleration.

6 Related Work

PTQ algorithms. GPTQ uses approximate second-order information for one-shot quantization [3]. AWQ protects salient channels using activation-aware statistics [7]. SmoothQuant migrates activation outlier difficulty into weights for W8A8 inference [14]. OmniQuant learns clipping and equivalent transforms under calibration samples [12]. These are backend families: they answer how to quantize once a representation or protection policy is chosen. CSI answers a different question before that policy is executed: whether a calibration-derived allocation or protection decision is stable enough to be trusted. Our full Qwen2.5-1.5B rows therefore confront AutoAWQ and GPTQModel as native PTQ baselines, while CSI is framed as a pre-allocation constraint layer that can gate such backends rather than as a drop-in replacement for them.

Rotation and outlier handling. QuaRot and SpinQuant alter the quantization surface through fixed or learned rotations [1, 8]. CSI is complementary: it asks whether calibration-dependent sensitivity or protection decisions are stable across plausible prompt samples.

Calibration and benchmarking. Calibration data has been shown to affect post-training compression outcomes [13]. Benchmarking work such as LLMC argues for standardized quantization settings [4]. CSI contributes a narrower pre-allocation audit: before arguing about final task scores, test whether the module ranking used for allocation is stable.

7 Limitations and Falsification

The strongest current evidence is a reproducible CSI measurement and gating pipeline plus matched native-PTQ downstream retention on full Qwen2.5-1.5B MMLU/GSM8K fixtures. Local GPU fake-quant smokes now provide directional allocation-path evidence on Qwen2.5-0.5B and a Qwen2.5-1.5B single-slice run. We therefore make a system-level audit claim rather than a broad PTQ SOTA claim: CSI identifies when calibration-derived allocation evidence is stable enough to trust, and it marks the exact evidence still needed before claiming end-to-end downstream superiority.

Before submission, the most important missing row is a direct CSI-informed downstream retention stress test in the 2–3 bit regime. The current artifacts contain full 1.5B MMLU/GSM8K native-PTQ retention and local 7B/14B feasibility, plus uniform INT3 and synthetic INT2/INT3 distortion

Table 4: Full downstream retention for Qwen2.5-1.5B under matched native PTQ task fixtures. Deltas are candidate-minus-FP16 paired bootstrap intervals.

Task	Variant	N	Acc.	Wilson CI	Δ vs FP16	Δ CI
MMLU	FP16	14042	0.5864	[0.5782, 0.5945]	–	–
MMLU	AutoAWQ	14042	0.5648	[0.5566, 0.5730]	-0.0216	[-0.0273, -0.0155]
MMLU	GPTQModel	14042	0.5362	[0.5279, 0.5444]	-0.0502	[-0.0570, -0.0432]
GSM8K	FP16	1319	0.0811	[0.0676, 0.0971]	–	–
GSM8K	AutoAWQ	1319	0.0788	[0.0655, 0.0946]	-0.0023	[-0.0190, 0.0144]
GSM8K	GPTQModel	1319	0.0728	[0.0600, 0.0881]	-0.0083	[-0.0243, 0.0076]

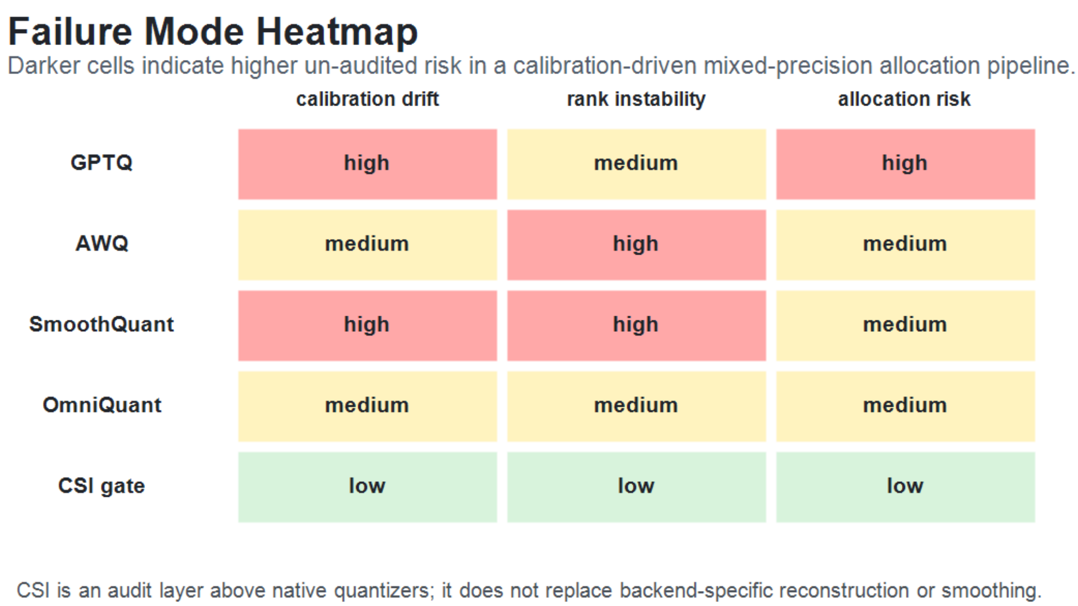

Figure 6: Failure mode heatmap across PTQ method families. Darker cells indicate higher un-audited risk in a calibration-driven allocation pipeline, not that a backend is weak in its native setting. CSI is complementary to GPTQ, AWQ, SmoothQuant, and OmniQuant: it audits the calibration/allocation decision above the native quantizer.

Table 5: Local GPU fake-quant PPL smoke. Lower is better. The consensus allocation uses about 4.5 average bits, so these rows are directional feasibility evidence rather than same-budget downstream claims.

Model	Slice	FP16	INT4	INT3	CSI alloc.
0.5B	WikiText2-8	24.73	41.50	915.60	32.78
0.5B	C4-8	30.64	47.26	1138.74	40.35
0.5B	WikiText2-16	25.01	42.23	804.83	32.34
0.5B	C4-16	28.85	44.86	962.40	38.21
1.5B	WikiText2-1	10.82	15.63	–	13.51

scaffolds, but they do not yet contain a full task-retention curve over 2, 3, 4, and 8 bits for FP16, uniform, native PTQ, and CSI-informed allocations. That experiment should be run before upgrading the paper from a robustness-plus-retention claim to a method-superiority claim.

Table 6: Mechanistic ablation map for the constrained formulation.

Removed part	Failure mode	Evidence hook
CSI gate	rank noise	$n = 2$ $\rho = 0.1410$
Consensus	bad split	$+2.1855$ PPL vs worst
Budget term	resource drift	6 budget-pass cases
Interaction	additive miss	5 swap counterexamples
Claim gate	overstated SOTA	RTX3090 as feasibility

Several results could weaken the story. A second public prompt pool could fail to reproduce the stability trend. Another model family could show stable rankings even at very small n. Downstream task retention could be insensitive to the measured ranking instability once CSI-informed allocations are directly evaluated. A faithful AWQ, GPTQ, SmoothQuant, OmniQuant, QuaRot, or SpinQuant implementation could

Table 7: Local RTX3090 guarded task rows.

Variant	Task	Limit	Acc.	Tok/s
7B FP16	MMLU	100	0.71	8.376
7B AWQ	MMLU	100	0.70	6.765
7B GPTQ	MMLU	100	0.74	7.231
7B FP16	GSM8K	100	0.24	15.263
7B AWQ	GSM8K	100	0.24	11.416
7B GPTQ	GSM8K	100	0.16	10.828
14B AWQ	MMLU	10	0.50	3.851

Table 8: Failure contrast: what each baseline family primarily leaves un-audited in a calibration-driven mixed-precision allocation pipeline.

Family	Strength	Unchecked failure
GPTQ	reconstruction backend	split drift
AWQ	salient-channel backend	ranking collapse
SmoothQuant	outlier-scaling backend	allocation drift
OmniQuant	learned-transform backend	gate conflict
CSI	decision boundary	retention gap

make identical allocation decisions across calibration splits. Finally, a simpler budget-matched heuristic could beat CSI-informed allocation once direct retention rows are available.

8 Conclusion

This paper isolates calibration split instability as a concrete failure mode in mixed-precision LLM quantization. Small calibration samples can produce unstable sensitivity rankings, and the audited artifacts show that increasing calibration size improves rank and set agreement under two evidence lines. CSI turns this observation into a reproducible decision operator: if the calibration ranking is unstable, the correct output is REJECT; if it is stable but lacks direct retention, the correct output is AUDITONLY; and only fixed-budget downstream retention can justify VALIDATE. Full Qwen2.5-1.5B MMLU/GSM8K rows show how this operator can be paired with native PTQ retention evidence rather than left as a standalone diagnostic. The immediate next step is direct downstream retention testing of CSI-informed allocations against strong native PTQ and calibration-aware baselines under matched memory budgets.

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